极客时间

递归、分治
（Recursion，Divide & Conquer）

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扫码了解极客时间《算法面试通关40讲》视频课程

本节内容

1. Recursion
2. Divide & Conquer

Recursion

递归 — 循环

通过函数体来进行的循环

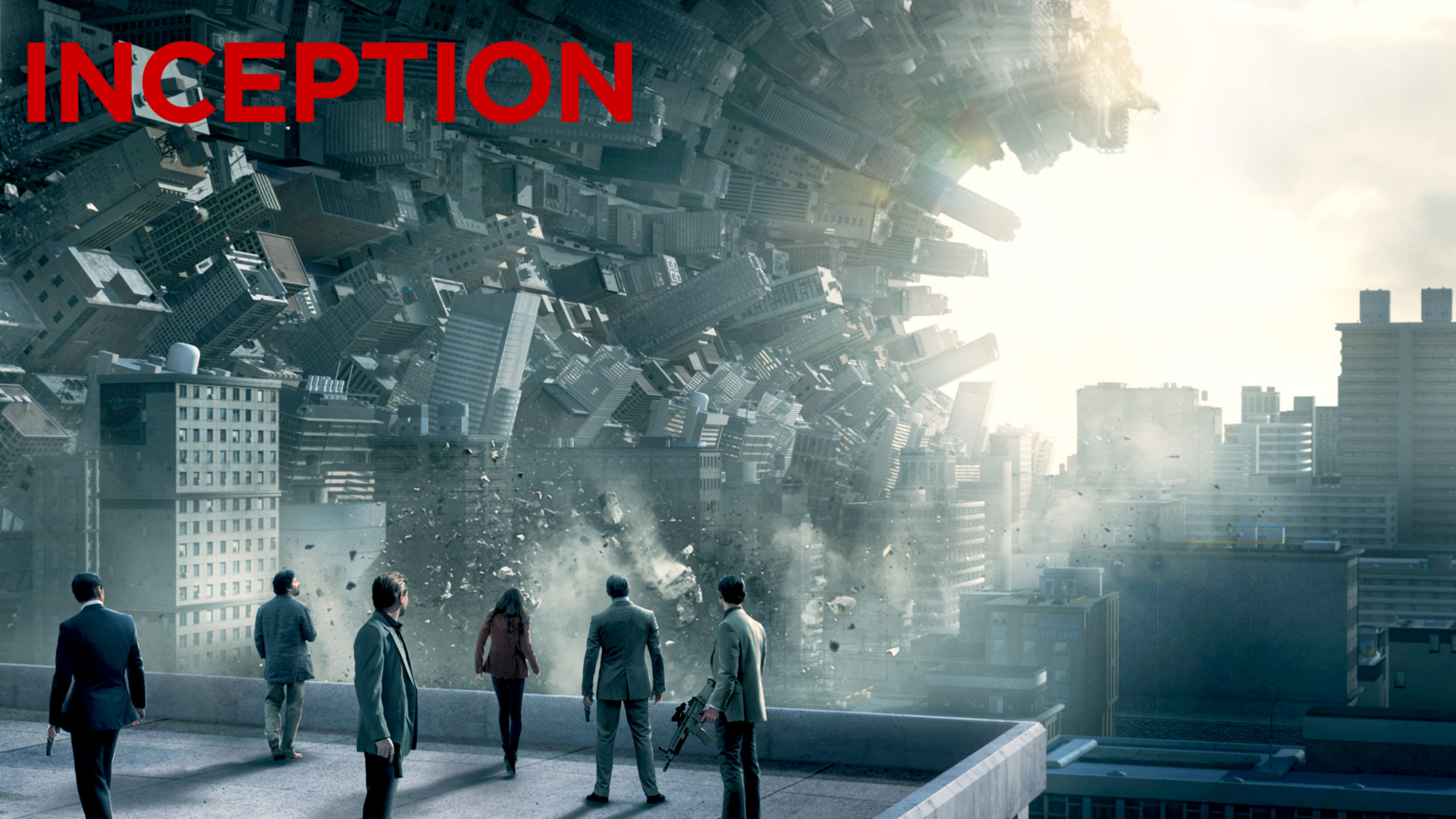
递归

1.从前有个山，

2.山里有个庙，

3.庙里有个和尚讲故事：

INCEPTION



Recursion

计算 n!

n! = 1 * 2 * 3 * … * n

```
def Factorial(n):
    if n <= 1:
        return 1
    return n * Factorial(n - 1)
```

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Recursive

```
factorial(6)
6 * factorial(5)
6 * (5 * factorial(4))
6 * (5 * (4 * factorial(3)))
6 * (5 * (4 * (3 * factorial(2))))
6 * (5 * (4 * (3 * (2 * factorial(1)))))
6 * (5 * (4 * (3 * (2 * 1))))
6 * (5 * (4 * (3 * 2)))
6 * (5 * (4 * 6))
6 * (5 * 24)
6 * 120
720
```

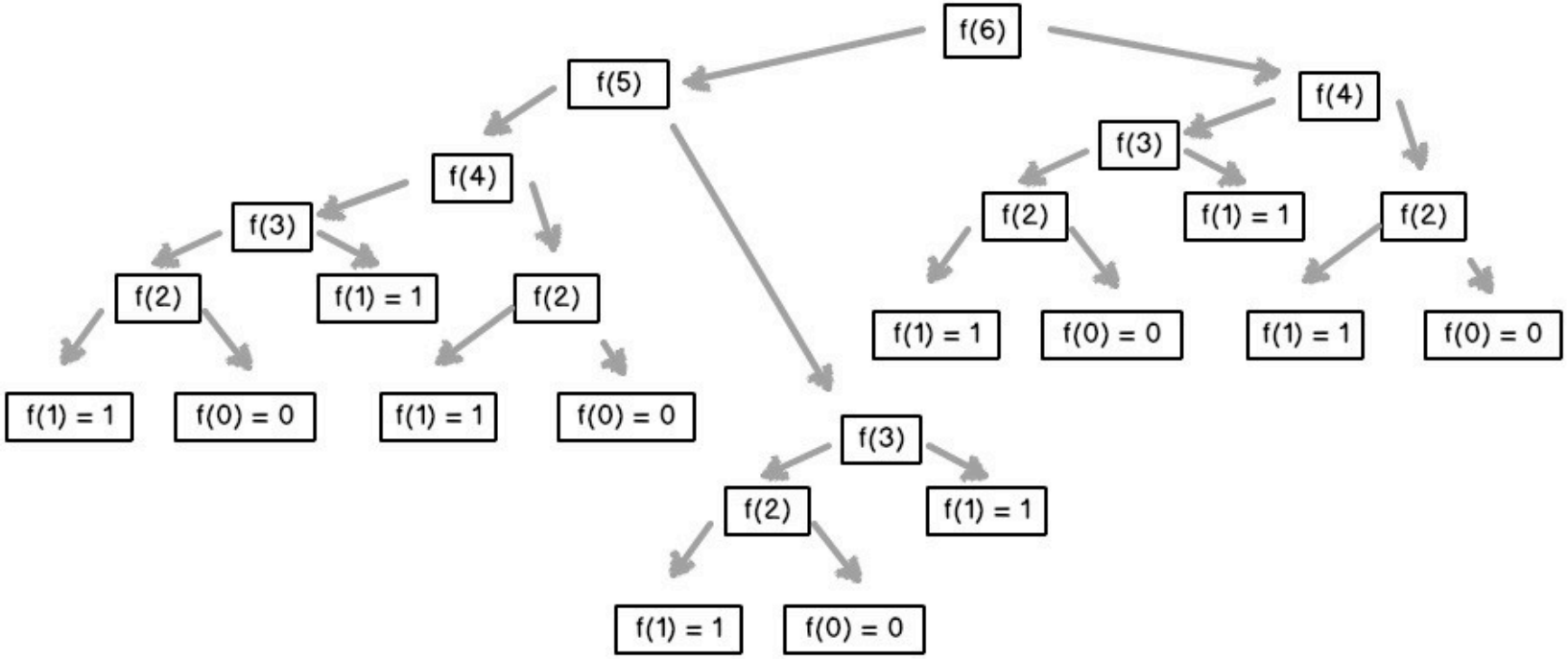
Recursion

Fibonacci array: 1, 1, 2, 3, 5, 13, 21, 34, …

F(n) = F(n-1) + F(n-2)

```
def fib(n):
    if n == 0 or n == 1:
        return n
    return fib(n - 1) + fib(n - 2)
```

Fib(6)


f(6)
f(5)
f(4)
f(4)
f(3)
f(3)
f(2)
f(1) = 1
f(2)
f(2)
f(1) = 1
f(2)
f(1) = 1
f(0) = 0
f(1) = 1
f(0) = 0
f(1) = 1
f(0) = 0
f(1) = 1
f(0) = 0
f(3)
f(2)
f(1) = 1
f(1) = 1
f(0) = 0

```python
def recursion(level, param1, param2, ...):

  # recursion terminator
  if level > MAX_LEVEL:
    print_result
    return

  # process logic in current level
  process_data(level, data...)

  # drill down
  self.recursion(level + 1, p1, ...)

  # reverse the current level status if needed
  reverse_state(level)
```

分治 - Divde & Conquer

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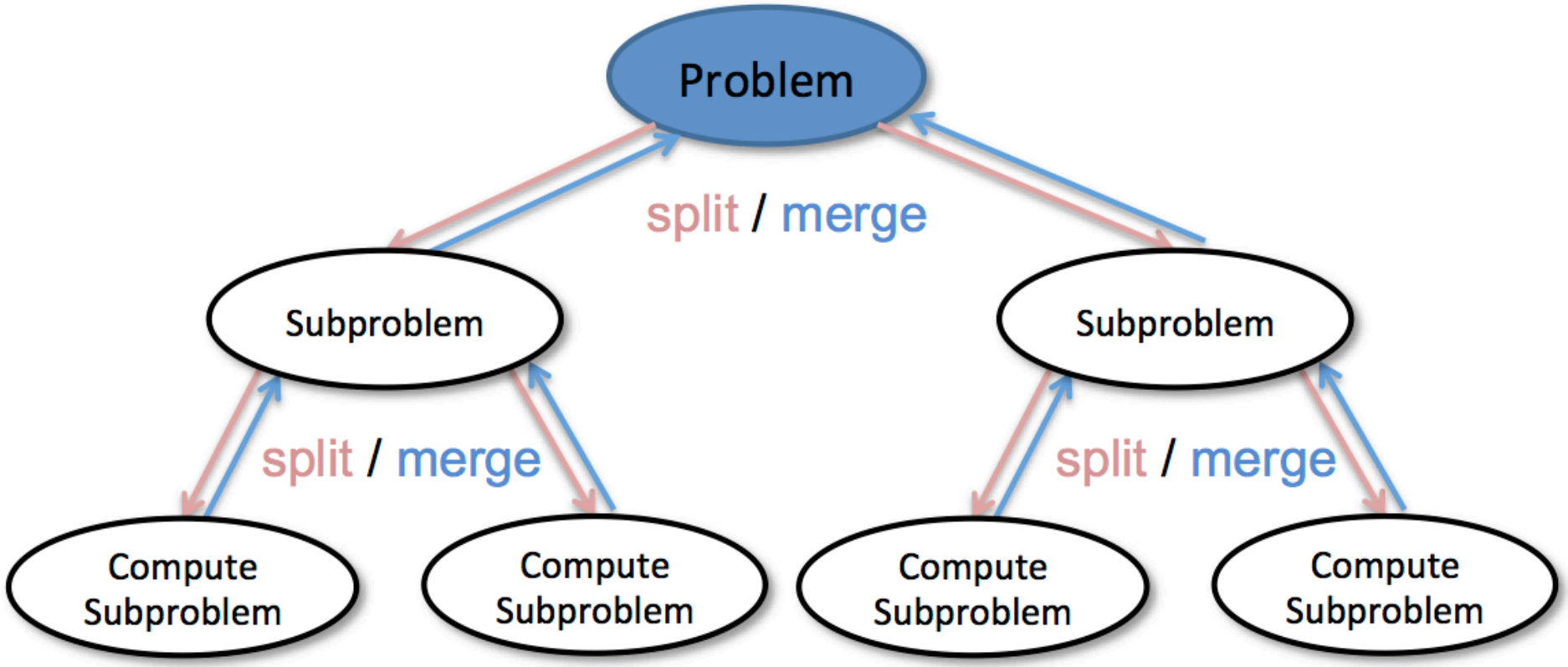
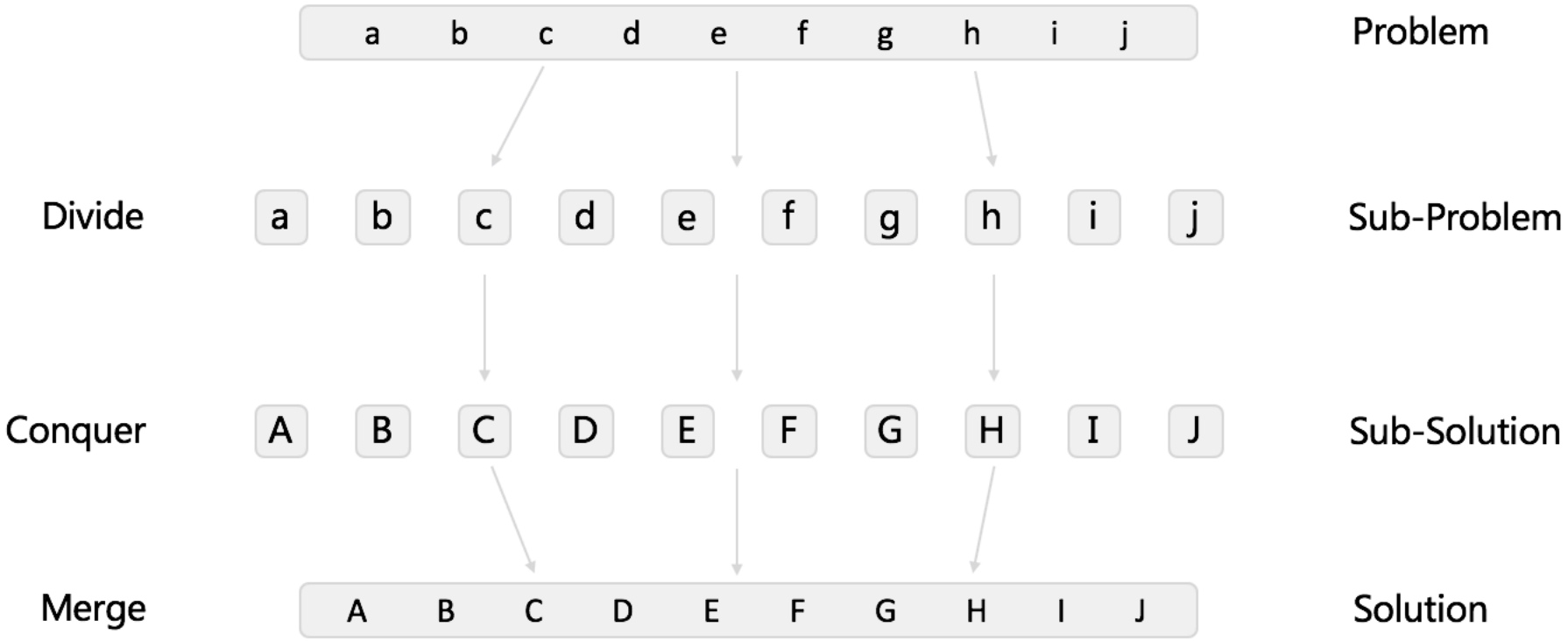

Problem
split / merge
Subproblem
Subproblem
split / merge
split / merge
Compute Subproblem
Compute Subproblem
Compute Subproblem
Compute Subproblem

Divide & Conquer



```python
def divide_conquer(problem, param1, param2, ...):

  # recursion terminator
  if problem is None:
    print_result
    return

  # prepare data
  data = prepare_data(problem)
  subproblems = split_problem(problem, data)

  # conquer subproblems
  subresult1 = self.divide_conquer(subproblems[0], p1, ...)
  subresult2 = self.divide_conquer(subproblems[1], p1, ...)
  subresult3 = self.divide_conquer(subproblems[2], p1, ...)
  ...

  # process and generate the final result
  result = process_result(subresult1, subresult2, subresult3, ...)
```

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实战题目

1. https://leetcode.com/problems/powx-n/description/
2. https://leetcode.com/problems/majority-element/description/
3. https://leetcode.com/problems/maximum-subarray/description/
4. https://leetcode.com/problems/valid-anagram/#/description
5. https://leetcode.com/problems/find-all-anagrams-in-a-string/#/description
6. https://leetcode.com/problems/anagrams/#/description

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